## BME451- APPLIED THERMODYNAMICS LAB

**Objectives:**

**CO1:** To understand the principles of various boilers: **K2**

**CO2:** To understand the basic principles IC engines and determination of various performance parameters of IC Engines: **K3**

**CO3:** To understand the principles of steam engine and Steam & Gas Turbine: **K2**

**List of Experiments: (At least 8 of the following)**

1. Study of Fire Tube boiler.
2. Study of Water Tube boiler.
3. Study and working of Two stroke petrol Engine.
4. Study and working of Four stroke petrol Engine.
5. Determination of Indicated H.P. of I.C. Engine by Morse Test.
6. Prepare the heat balance sheet for Diesel Engine test rig.
7. Prepare the heat balance sheet for Petrol Engine test rig.
8. Study and working of two stroke Diesel Engine.
9. Study and working of four stroke Diesel Engine.
10. Study of Velocity compounded steam turbine.
11. Study of Pressure compounded steam turbine.
12. Study of Impulse & Reaction turbine.
13. Study of steam Engine model.
14. Study of Gas Turbine Model.

## BME452- MANUFACTURING PROCESSES LAB

**CO-1** : Students will be able to make the component using casting and finishing methods. **K2**

**CO-2 :** Students will be able to make the component using metal cutting / unconventional machining methods. **K2**

**CO-3 :** Students will be able to make the component using metal joining processes. **K2**

**List of Experiments: (At least 8 of the following along-with study of the machines/processes)**

1. Making of Pattern (Wax / Wooden)
2. Preparation of Mould and Casting
3. Shear-angle determination (using formula) with tube cutting (for orthogonal) on lathe machine.
4. Bolt (thread) making on Lathe machine.
5. Tool grinding (to provide tool angles) on tool-grinder machine.
6. Gear cutting on Milling machine.
7. Finishing of a surface on surface-grinding machine.
8. Drilling holes on drilling machine and study of twist-drill.
9. Study of different types of tools and its angles & materials.
10. Experiment on tool wear and tool life.
11. Experiment on jigs/Fixtures and its uses.
12. Gas welding experiment.
13. Arc welding experiment.
14. Resistance welding experiment.
15. Soldering & Brazing experiment.
16. Experiment on Unconventional Machining (any one among - Laser Cutting, CO2 Cutting, ECM, EDM etc.)

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